

Soil Health & Crop Recommendation Report

Report Date: November 07, 2025 at 20:16:28 UTC

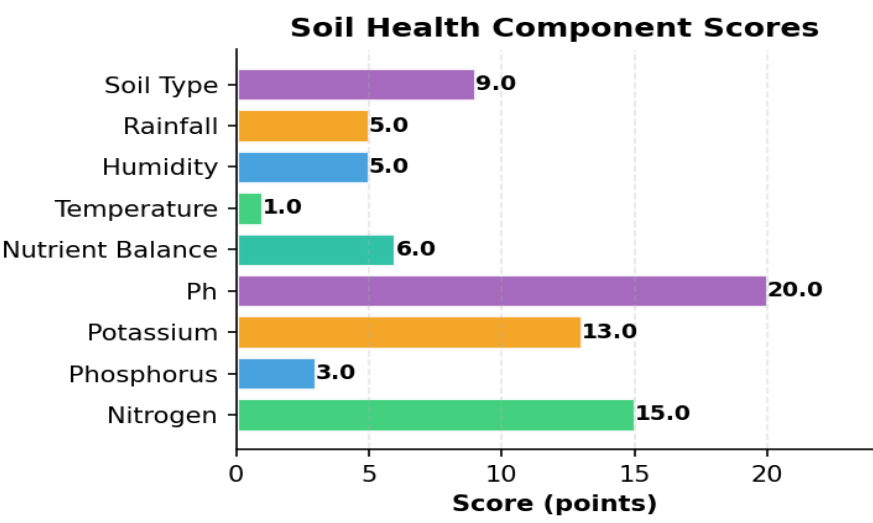
Farmer Information	
Name:	ravi bhosle
Phone:	8088117817
Location:	tonshyal
Land Size:	21.00 acres

Parameter	Value	Unit
Nitrogen (N)	75.0	mg/kg
Phosphorus (P)	102.0	mg/kg
Potassium (K)	79.0	mg/kg
pH Level	6.50	
Temperature	35.3	°C
Humidity	67.5	%
Rainfall	120.0	mm
Soil Type	Black Soil	

Soil Health Assessment

Overall Soil Health Score	77.0/100	Good	
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Component	Score	Points
Nitrogen	15.0	points
Phosphorus	3.0	points
Potassium	13.0	points
Ph	20.0	points
Nutrient Balance	6.0	points
Temperature	1.0	points
Humidity	5.0	points
Rainfall	5.0	points
Soil Type	9.0	points



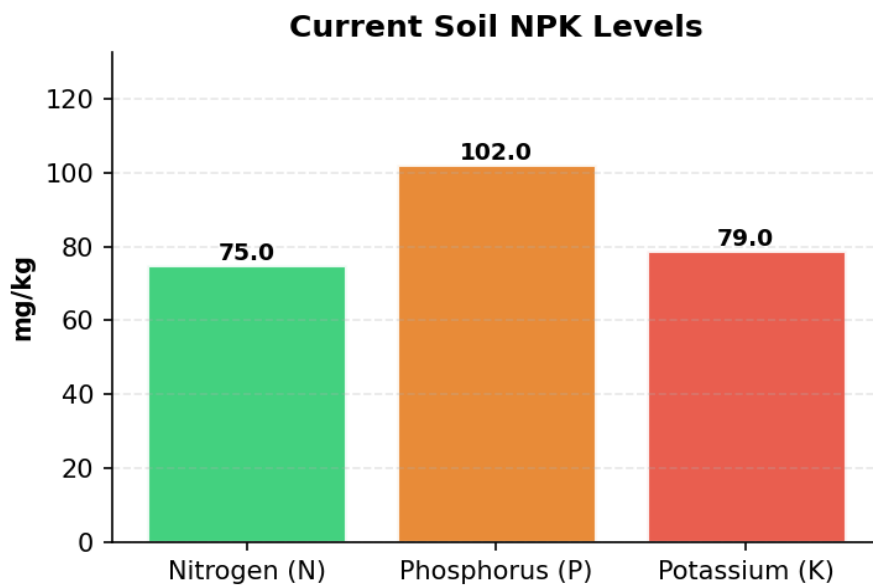
Recommended Crop

Predicted Crop	Banana
Confidence Level	24.3%

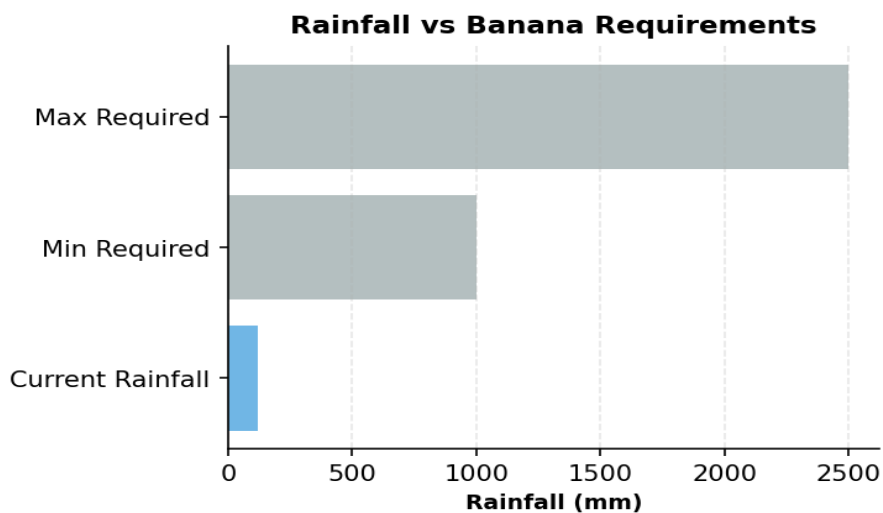
Alternative Crop Options

Rank	Crop	Confidence (%)
1	Pigeonpeas	17.1%
2	Cotton	13.8%

Soil NPK Levels Visualization



Rainfall Analysis



Important Warnings

- Very high phosphorus levels detected. May interfere with micronutrient uptake.

Soil Treatment Recommendations

- Soil pH (6.5) is within optimal range (6.0-7.5) for banana.
- Nitrogen (75.0 mg/kg) is low for banana. Optimal range: 180-220. Add compost, green manure, or urea (125 kg/acre needed).
- Phosphorus (102.0 mg/kg) is excessive for banana. Optimal range: 40-60. High P can interfere with micronutrient uptake.

- Potassium (79.0 mg/kg) is low for banana. Optimal range: 280-320. Add wood ash, banana peel powder, or potash (221 kg/acre needed).
- High clay content, retains moisture well. Ideal for cotton, wheat, and pulses.

Fertilizer Recommendations

Fertilizer Requirements for banana:

- Land Size: 21.0 acres
- Current NPK: N=75.0, P=102.0, K=79.0 mg/kg
- Ideal NPK: N=200, P=50, K=300 kg/acre

Required Application:

- Nitrogen (N): 2625.0 kg total (125.0 kg/acre)
→ Apply in 2-3 split doses: 50% at sowing, 25% at tillering, 25% at flowering
- Potassium (K): 4641.0 kg total (221.0 kg/acre)
→ Apply in 2 splits: 50% basal, 50% at flowering

Organic Alternatives:

- Compost/FYM: 5-10 tons/acre (provides N, P, K)
- Green Manure: Leguminous crops (fixes N)
- Biofertilizers: Rhizobium (N), PSB (P), VAM (P)

Crop-Specific Tips for Banana

Requires high humidity and consistent moisture. Plant in pits with organic matter.

Additional Notes

- These recommendations are based on comprehensive soil analysis and advanced ML predictions.
- Actual results may vary based on local conditions, weather patterns, and farming practices.
- Consult with local agricultural experts before making major farming decisions.
- Regular soil testing (every 2-3 years) is recommended for optimal crop management.
- Consider crop rotation and organic farming practices for long-term soil health.
- Monitor weather forecasts and adjust irrigation schedules accordingly.

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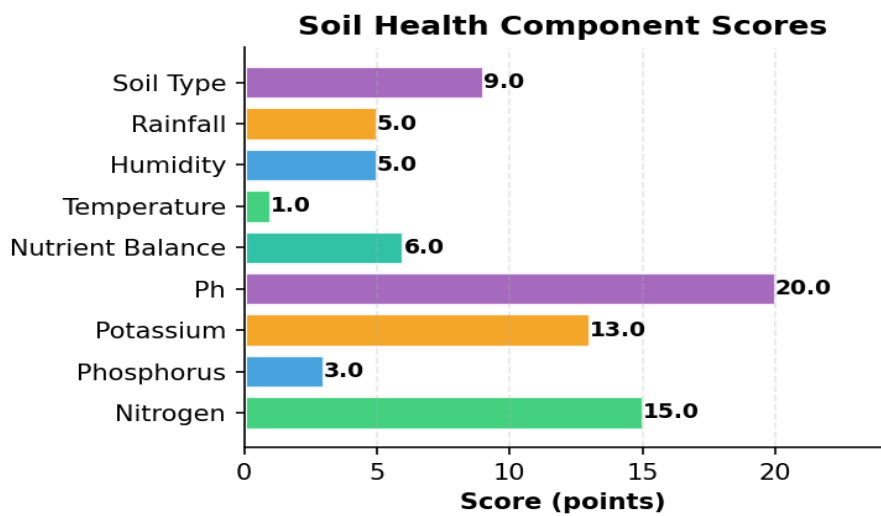
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Soil Health & Crop Recommendation System

Comprehensive Agricultural Analysis Report



Recommended Crop

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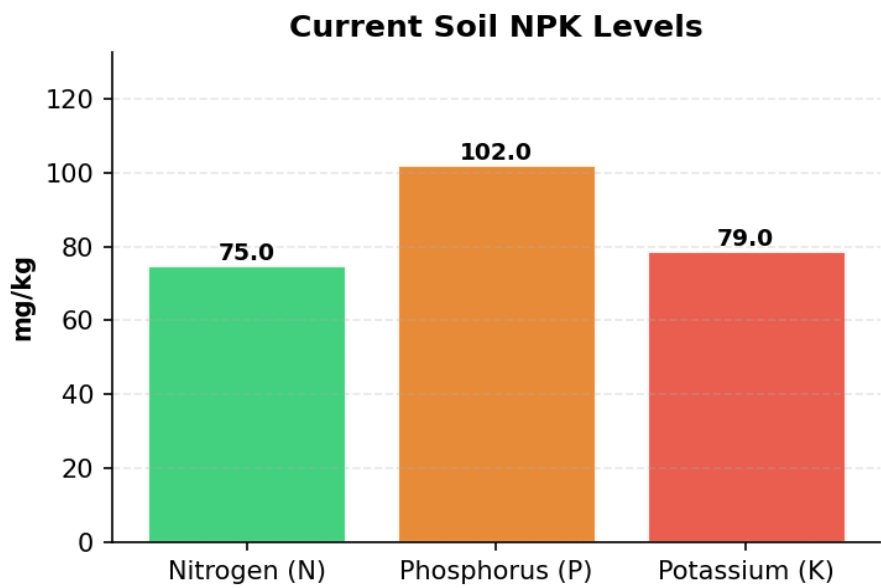
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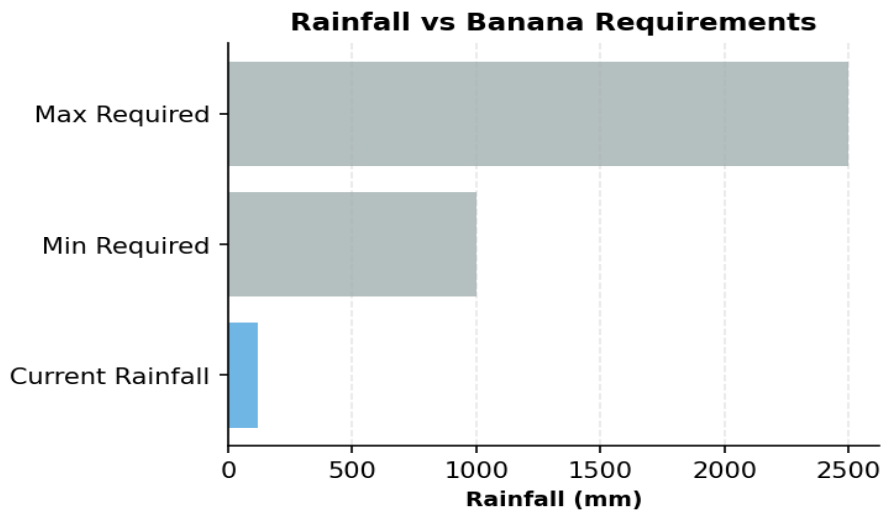
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